

COSC 39: Theory of Computation

Credit Statement: Talked to Carson Bates'27.

Problem 1. Prove that there is no Turing machine that decides the language:

1. $\text{REJECTSELF} := \{\langle M \rangle \mid M \text{ is a Turing machine which rejects } \langle M \rangle\}.$
2. $\text{NOTACCEPTSELF} := \{\langle M \rangle \mid M \text{ is a Turing machine that does not accept } \langle M \rangle\}.$

Solution.

1. Assume for contradiction that some Turing machine M_R decides

$$\text{REJECTSELF} := \{\langle M \rangle \mid M \text{ is a Turing machine which rejects } \langle M \rangle\}.$$

Construct a wrapper machine REJECTOR around M_R as follows: on input $\langle M \rangle$, REJECTOR runs M_R on $\langle M \rangle$.

- If M_R accepts $\langle M \rangle$, then REJECTOR accepts and halts.
- If M_R rejects $\langle M \rangle$, then REJECTOR rejects and halts.

To derive a contradiction, feed the encoding $\langle \text{REJECTOR} \rangle$ into REJECTOR itself.

- If REJECTOR accepts and halts $\langle \text{REJECTOR} \rangle$, then M_R must have accepted $\langle \text{REJECTOR} \rangle$. Since M_R decides REJECTSELF, this implies that REJECTSELF rejects $\langle \text{REJECTOR} \rangle$, a contradiction.
- If REJECTOR rejects and halts on input $\langle \text{REJECTOR} \rangle$, then M_R must have rejected $\langle \text{REJECTOR} \rangle$. Since M_R decides REJECTSELF, this implies that REJECTOR accepts $\langle \text{REJECTOR} \rangle$, a contradiction.

In both cases we obtain a contradiction. Therefore, no such Turing machine M_R exists, and REJECTSELF is undecidable.

2. Assume for contradiction that some Turing machine M_N decides

$$\text{NOTACCEPTSELF} := \{\langle M \rangle \mid M \text{ is a Turing machine that does not accept } \langle M \rangle\}.$$

Construct a wrapper machine NOT around M_N as follows: on input $\langle M \rangle$, NOT runs M_N on $\langle M \rangle$.

- If M_N accepts $\langle M \rangle$, then NOT accepts and halts.
- If M_N rejects $\langle M \rangle$, then NOT rejects and halts.

To derive a contradiction, feed the encoding $\langle \text{NOT} \rangle$ into NOT itself.

- If NOT accepts and halts $\langle \text{NOT} \rangle$, then M_N must have accepted $\langle \text{NOT} \rangle$. Since M_N decides NOTACCEPTSELF, this implies that NOT rejects $\langle \text{NOT} \rangle$, a contradiction.
- If NOT rejects and halts on input $\langle \text{NOT} \rangle$, then M_N must have rejected $\langle \text{NOT} \rangle$. Since M_N decides NOTACCEPTSELF, this implies that NOT accepts $\langle \text{NOT} \rangle$, a contradiction.

In both cases we obtain a contradiction. Therefore, no such Turing machine M_N exists, and NOTACCEPTSELF is undecidable.